----- Active space -----

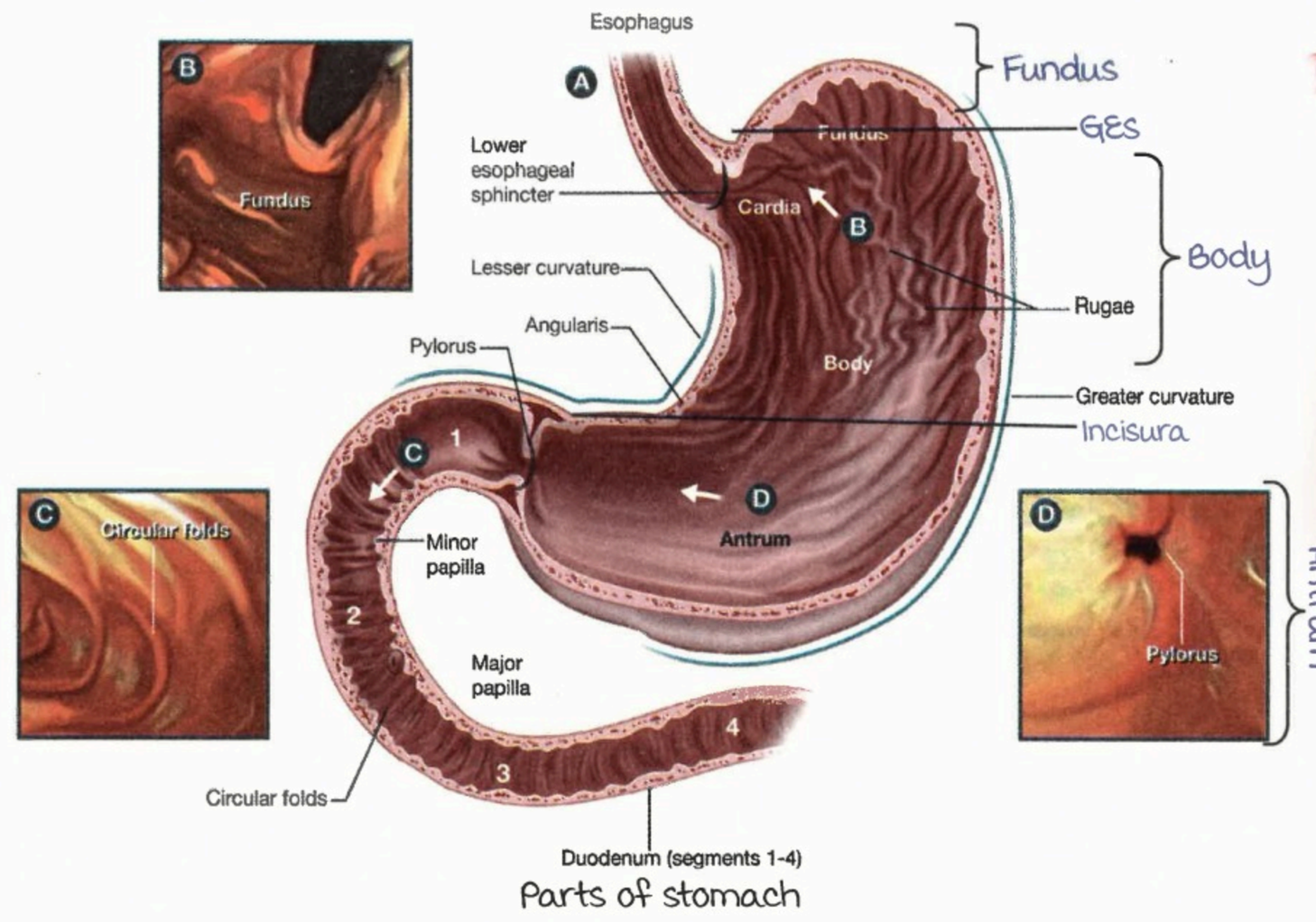
STOMACH

Anatomy and Physiology

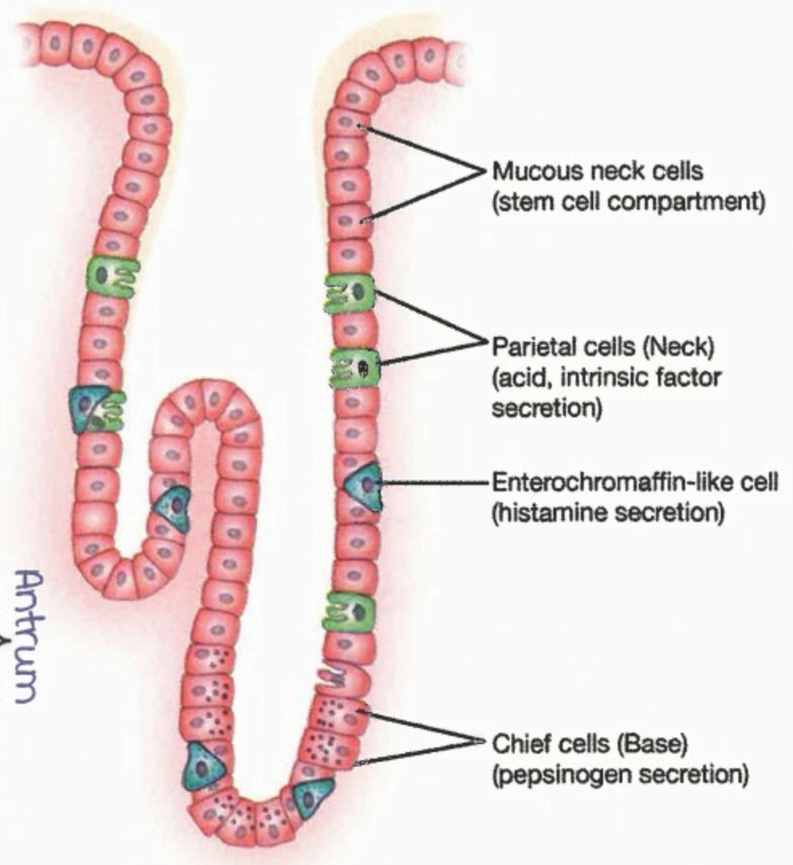
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Parts of stomach :

	Fundus	Body/corpus	Antrum
Location	Above gastro-Esophageal sphincter (GES)	B/w GES & incisura	Below incisura
Glands	Fundic glands	Oxyntic glands	Pyloric glands
Cells	• Entero chromaffin Like cells (ECL). • Mucous cells. • Endocrine cells.	• Parietal cells (neck/ isthmus). • Chief cells (base). • Mucus cell. • ECL.	• G cells. • Mucous cell. • ECL.
Secretion	ECL → Histamine	Parietal cell → Acid, intrinsic factor. Chief cells → Pepsinogen	G cells → Gastrin



Parts of stomach



Oxyntic glands

Layers of gastroduodenal mucosal defence :

- Pre epithelial layer.
- Epithelial layer : Restitution.
- Sub epithelial layer.

Damaging factors :

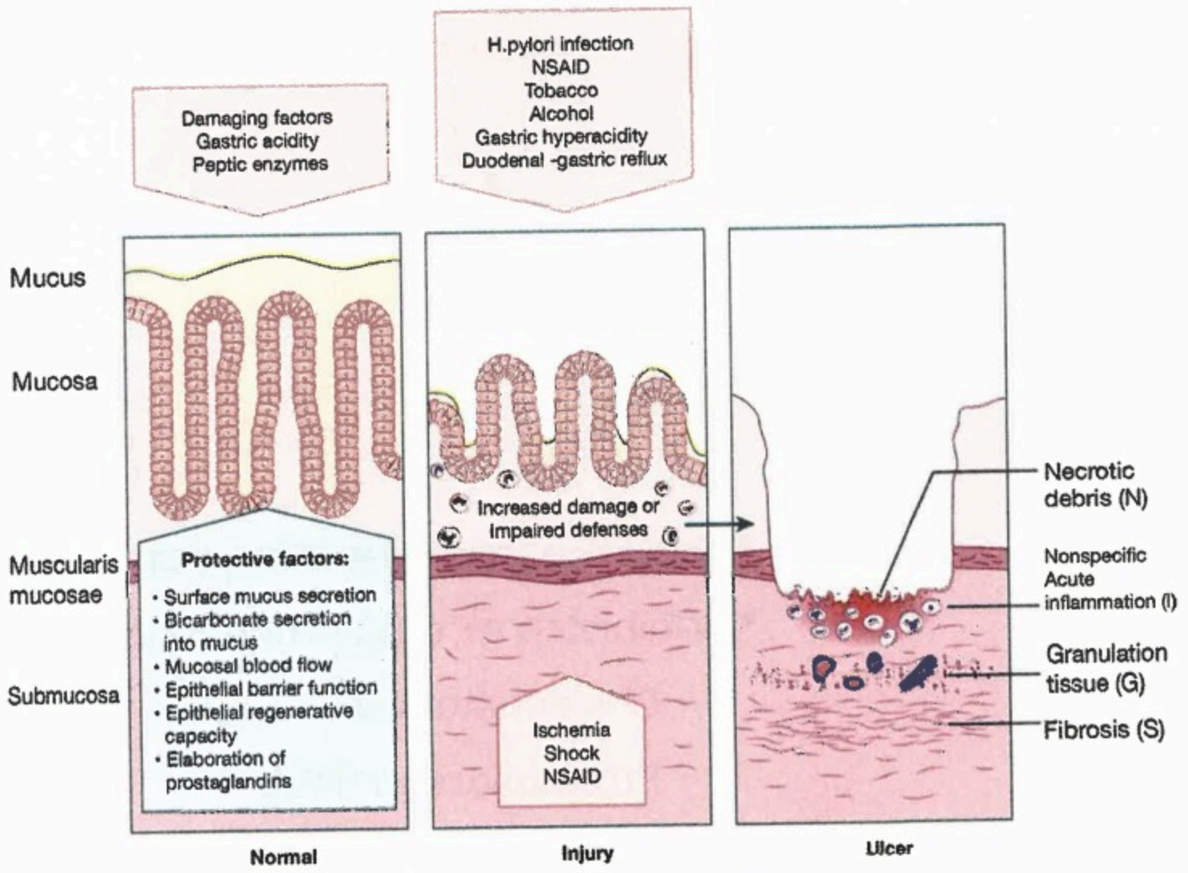
- Acid.
- NSAIDs.
- Tobacco.
- H. Pylori (Not always).

Protective factors :

COX-1.

↓

Prostaglandins (PG). } in epithelial layer



mechanism of Gastric injury and protection

Layers of GIT :

- Mucosa :
  - Mucosal gel coating.
  - Epithelial layer.
  - Lamina propria.
  - Muscularis mucosa.
- Serosa.
- Submucosa : meissner's plexus.
- muscularis layer :
  - Outer longitudinal.
  - Auerbach's plexus.
  - Inner circular.

Note :

Pacemaker cells of GIT : Interstitial cells of Cajal

↓

Produce basal electric rhythm (BER)

Enteric nervous system (ENS) :

Intrinsic innervation

↓

- Auerbach's plexus : motility
- meissner's plexus : Intestinal secretion

Extrinsic innervation

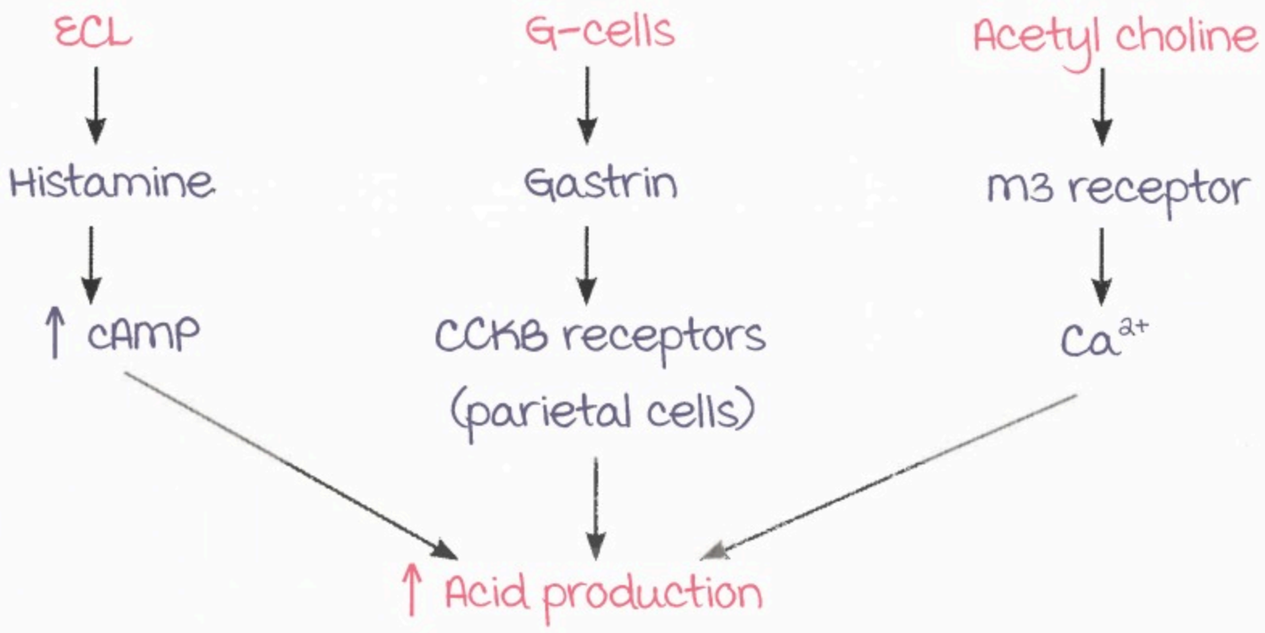
↓

Autonomic Nervous System (ANS)

- sympathetic : ↓ motility
- parasympathetic : ↑ motility

----- Active space -----

Co-operative synergism :



- GRP (Gastrin Releasing Peptide)/Bombesin ⟶ ↑ Gastrin
- Inhibitors of acid production :
  - Cholecystokinin (CCK).
  - Prostaglandin (PG).
  - Somatostatin.
  - Gastric Inhibitory Peptide (GIP).
- Basal Acid output (BAO) :
  - Maximal acid output (MAO) : 6 pm.
  - Least acid output : 6 am.
  - BAO/MAO > 0.6 : Gastrin secreting tumour.

Gastritis

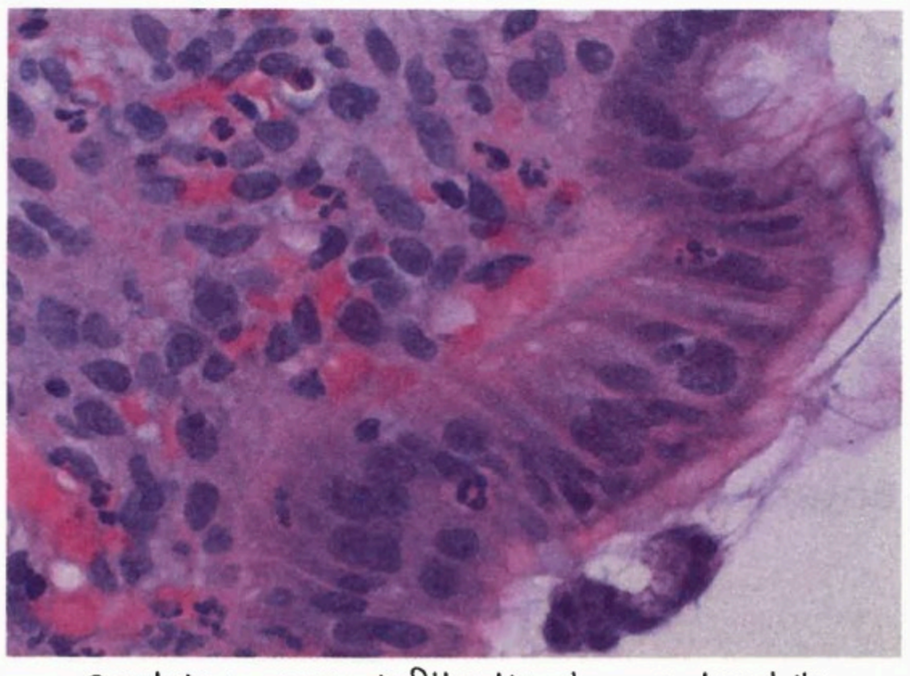
00:10:06

Definition : Histological inflammation of gastric mucosa.

Causes : NSAIDs > infectious.

Diagnosis :

- Clinical.
- Endoscopy & histology ⟶ No correlation.



Gastric mucosa infiltration by neutrophils

ACUTE PHLEGMONOUS GASTRITIS

Rare, severely progressive inflammation of gastric wall.
Thickening of wall and gas formation.

Seen in : Immunocompromised people.

Causative organism : Streptococci > E.coli.

Clinical features : Abdominal pain.
Vomiting.
Fever.

Complication : Stomach wall necrosis.
High mortality rate.

CHRONIC GASTRITIS

----- Active space -----

Superficial involvement : Lymphocyte & plasma cells.

Types :

	Type A (less common)	Type B (more common)
AKA	Autoimmune metaplastic Atrophic Gastritis (AMAG)	Environmental metaplastic Atrophic Gastritis (EMAG)
Pattern of atrophy	Patchy : mainly corpus	Multifocal
Cause	Autoimmune : HLA B8/DR3	H. pylori
Complications	Antibody against parietal cells (body) ↓ Acid → Achlorhydria → Hypergastrinemia → Gastrin secreting carcinoid tumor ↓ Intrinsic factor (IF) → megaloblastic anemia	Gastritis → Gastric atrophy → Intestinal metaplasia → Gastric adenocarcinoma
Risk of malignancy	↓	↑

Note :

- Eosinophilic gastritis : Intestinal obstruction.
- Russel body gastritis : Pseudo tumour endoscopy appearance.
- Granulomatous gastritis : Crohn's disease (m/c cause).
- Lymphocytic gastritis : Celiac disease.

HYPERTROPHIC GASTRITIS

AKA menetrier's disease/protein losing gastropathy.

- Involve body and fundus (spare antrum) : Large tortuous mucosal folds.
- M > F.
- 40–60 yrs.
- Children : Associated CMV infection.

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Pathogenesis :

- Main cytokine : TGF-α.
- Foveolar hyperplasia

↓

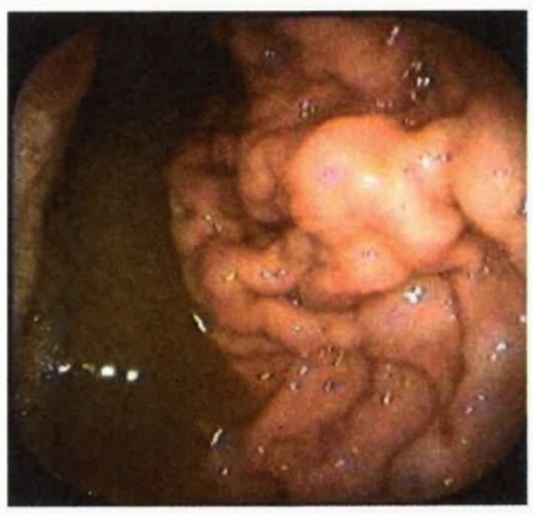
Hypersecretion of mucus

↓

Protein loss

Clinical features :

- Hypoalbuminenia.
- Edema.
- Upper GI symptoms.



Hypertrophic gastritis (endoscopy)

Investigation : Endoscopy → Full thickness biopsy (as premalignant).

Treatment :
- Surgical (main).
- Medical : EGFR antagonist (Cetuximab).

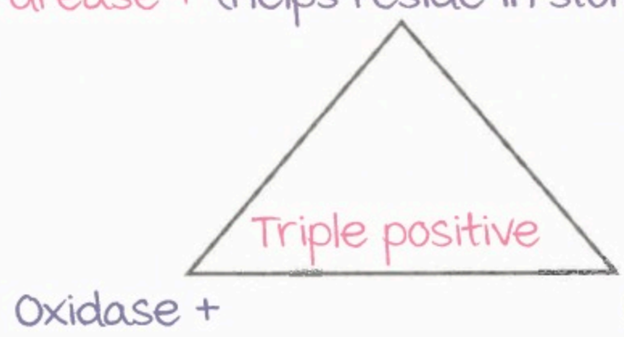
Helicobacter Pylori

00:19:30

INTRODUCTION

- Gram negative bacilli

Urease + (helps reside in stomach)



- Present in the antrum → Can migrate proximally (D/t flagella).
- Not invasive.
- Resides in the mucosal gel (deep).

PATHOLOGY

Transmission : Feco-oral → Colonisation depends on socio economic factors.

Progression : Colonised population (80% in developing countries)
- Majority : Chronic superficial gastritis
- 10% peptic ulcer

Enzymes produced
- Urease (creates alkaline environment).
- Protease (digest mucin).

Protective against :

- GERD.
- Barrett's oesophagus.
- Adenocarcinoma of oesophagus.

Risk factor for :

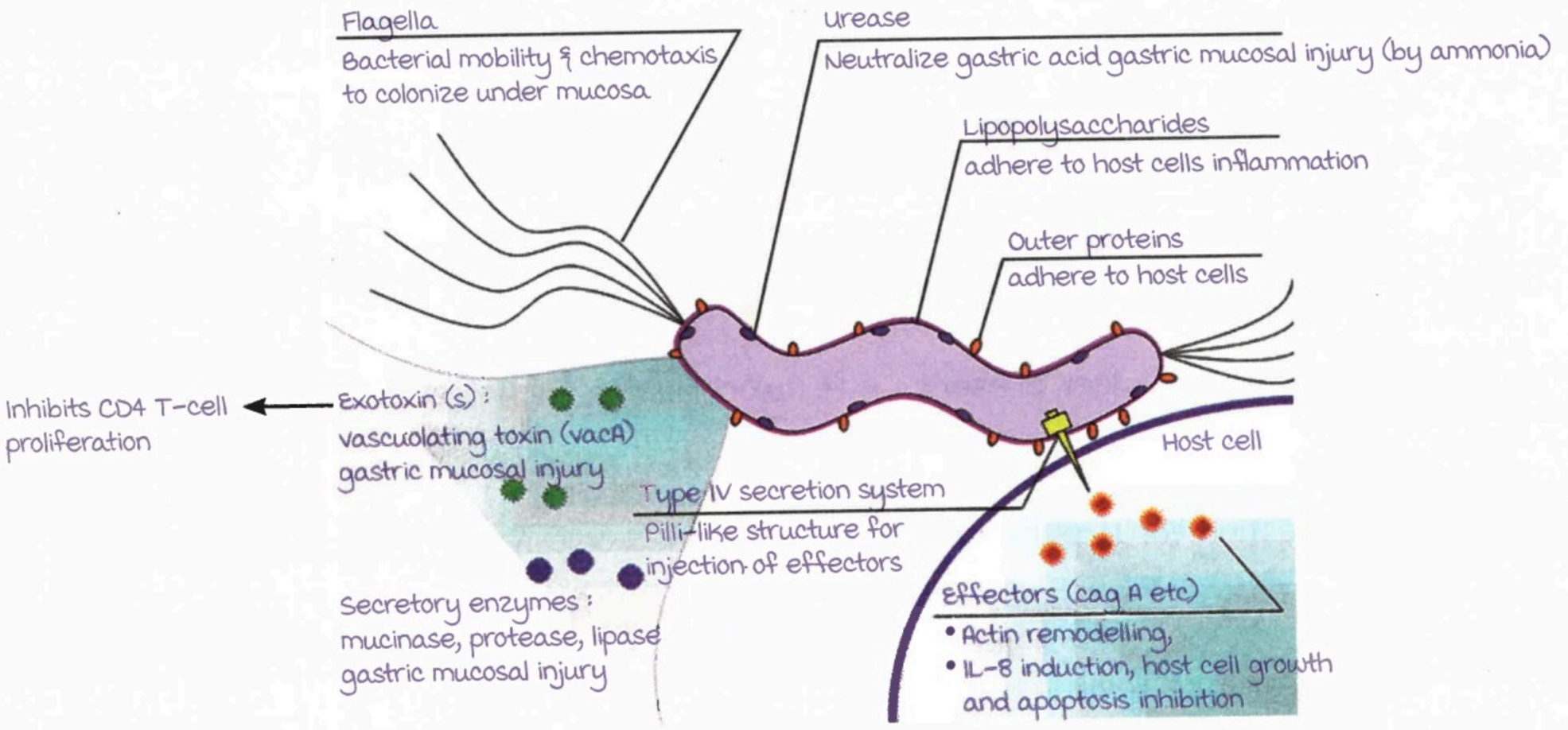
Adenocarcinoma of stomach.

Diseases caused :

----- Active space -----

- 70-80% of duodenal ulcers.
- 50-60% of gastric ulcers.
- Ca stomach.
- maltoma.

Virulence factors :



FORMS OF GASTRITIS

1. Antral predominant gastritis → Duodendal ulcer.
2. Corpus predominant atrophic gastritis → Gastric ulcer
   → Intestinal metaplasia → Dysplasia → Carcinoma.
3. Non atrophic pan-gastritis → Maltoma : m/c extranodal marginal zone lymphoma (B-cell).

Dyspepsia :

75% functional.

Symptoms :

- Post prandial fullness.
- Early satiety.
- Epigastric discomfort.

} Clinical diagnosis

management :

- 2 week trial of PPI (Daily dose).
- Indications for endoscopy :
  - >50 yrs.
  - Dysphagia, odynophagia.
  - Anaemia.
  - Weight loss.
  - Mass per abdomen.
  - H/o carcinoma stomach in family.

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INVESTIGATIONS

To confirm diagnosis :

1. Endoscopy :

Indication : Epigastric pain (Symptom of peptic ulcer disease).



Endoscopy → If ulcer present → H. Pylori testing → Anti H. Pylori Rx.

Note : ACG recommendation

Testing for H. pylori should be performed only if the clinician plans to offer treatment for positive results.

2. Rapid urease test :

Endoscopy → Antral biopsy → Urease test on tissue sample → Positive test : Change in colour.

To confirm eradication :

1. Stool antigen test.
2. C_{13}/C_{14} urea breath test.

Stool antigen test :

Sensitivity : 94%, Specificity : 97%.

Sensitivity is affected by : PPI, bismuth, antibiotics

↓

1 month gap after Rx, before performing test

TREATMENT

Drug therapy :

First line Rx : 14 day regimen of OCA/OCM

- Omeprazole : 20 mg BD.
- Clarithromycin : 500 mg BD.
- Amoxicillin : 1 g BD.
- metronidazole : 500 mg BD.

Second line Rx : OBTM

- Omeprazole.
- Bismuth sub salicylate.
- Tetracycline.
- metronidazole.

Note : Other 2nd line regimens are also used depending on the antibiotic resistance of the area.

Approach to management :

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